The empirical recurrence for OEIS A264268: recovering a conjecture that is not written down

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Abstract

OEIS A264268 records "Empirical recurrence of order 52", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 52, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 52 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A264268 is "Number of (n+1)X(4+1) arrays of permutations of 0..n*5+4 with each element having index change +-(.,.) 0,0 1,0 or 1,2.". It has offset 1 and begins

$$65, 1280, 40626, 1143000, 33729146, 981994496, 28735927165, 839596650695, \ldots$$

The entry states:

> Empirical recurrence of order 52 (see link above)

As of the "Last modified" line on the live entry (Nov 10 2015 11:30:52, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 52 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{52} c_i \, a(n-i),$$

c_1	24	c_{14}	389326578687	c_{27}	−336543397331056	c_{40}	−47984881699
c_2	411	c_{15}	−2125360577000	c_{28}	−46520765844531	c_{41}	21721475392
c_3	−6360	c_{16}	−2079876285629	c_{29}	241235675104680	c_{42}	3729595154
c_4	−61196	c_{17}	11735434374432	c_{30}	34232099835285	c_{43}	−1144491264
c_5	672824	c_{18}	7536887092416	c_{31}	−123769506884792	c_{44}	−172245969
c_6	4437625	c_{19}	−45291488450272	c_{32}	−18978490324031	c_{45}	36769704
c_7	−36769704	c_{20}	−18978490324031	c_{33}	45291488450272	c_{46}	4437625
c_8	−172245969	c_{21}	123769506884792	c_{34}	7536887092416	c_{47}	−672824
c_9	1144491264	c_{22}	34232099835285	c_{35}	−11735434374432	c_{48}	−61196
c_{10}	3729595154	c_{23}	−241235675104680	c_{36}	−2079876285629	c_{49}	6360
c_{11}	−21721475392	c_{24}	−46520765844531	c_{37}	2125360577000	c_{50}	411
c_{12}	−47984881699	c_{25}	336543397331056	c_{38}	389326578687	c_{51}	−24
c_{13}	263551716600	c_{26}	50928885196972	c_{39}	−263551716600	c_{52}	−1

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 52, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $52 + 4$ terms removed returns order 52 as well, so $\deg q_{\min} = 52 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $52+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 52 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A264268.

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